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Implementing mobile mammography in primary care: a community-based Reach, Effectiveness, Adoption, Implementation, and Maintenance (RE-AIM) evaluation

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Abstract

Background: **Mobile mammography units (MMU) address disparities in breast cancer screening. Challenges integrating MMU into routine healthcare delivery limit their sustained impact, particularly in underserved communities.**

Purpose: **This study evaluated the implementation of a mobile mammography program embedded within primary care practices described within the Practical, Robust Implementation and Sustainability Model (PRISM) and evaluated according to the Reach, Effectiveness, Adoption, Implementation, and Maintenance (RE-AIM) framework.**

Methods: **We conducted a REAIM implementation evaluation of a MMU program delivered from 2013 to 2019 within six primary care practices serving medically underserved populations. Data sources included administrative records, Patient Ambassador outreach and screening day logs, provider focus groups, and patient satisfaction surveys. RE-AIM domains were used to assess organizational uptake and sustainability, implementation processes, patient reach, and patient-reported experience.**

Results: **Practices adopted MMU integration with varied duration and intensity. Predictable screening schedules and clinic-based appointment prescheduling achieved greater screening completion and sustainability than sites relying primarily on patient outreach. Patient Ambassadors facilitated implementation through scheduling, logistics, and patient education, reducing administrative burden on clinical staff. Key implementation challenges included limited MMU capacity, missed appointments, staffing demands, and MMU accessibility constraints. Reach included over 6400 completed mammograms and more than 10 000 outreach calls. Patient satisfaction and intention to use the MMU again remained high throughout the program.**

Conclusions: **Embedding MMU within primary care, supported by Patient Ambassadors, can expand access to breast cancer screening in underserved communities. Sustainable implementation requires predictable scheduling, adequate coordination capacity, leadership engagement, and adaptation to local clinical and patient contexts.**

Keywords mobile mammography, breast cancer screening, community-based interventions, primary care, RE-AIM, health equity

Lay Summary

Many women living in communities with few resources are not screened regularly for breast cancer because of lack of transportation, cost, lack of time off from work, and difficulty making appointments. Mobile mammography units, large vehicles equipped to provide mammograms, can help reduce some of these barriers, but they are often offered as standalone screening options. This paper describes a community-based program that linked mobile mammography units to primary care practices serving low-income women in Western New York. The program included clinics, community organizations, and Patient Ambassadors who helped with outreach, making appointments, education, and support on screening days. We wanted to learn how the program was adopted, implemented, and sustained over time. We also report

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how many women were reached and if patients were satisfied with the service. Linking mobile mammography units to primary care practices on a regular schedule increased access to screening. Patient Ambassadors were important to the practices offering the program and helped patients feel comfortable with screening. Challenges such as limited number of appointments, missed appointments, staffing needs, and accessibility issues remained. Linking mobile screening to primary care and offering community resources and peer support, can improve access to breast cancer screening and identify what is needed to sustain such programs over time.

Implications

Practice: Primary care practices seeking to expand breast cancer screening in underserved communities could prioritize predictable onsite mobile mammography scheduling and dedicated coordination capacity, supported by peer Patient Ambassadors, rather than relying primarily on outreach alone.
Policy: Policymakers and healthcare funders should support sustained financing and infrastructure for embedding mobile mammography and peer support roles within primary care settings to promote equitable access and long-term program viability.
Research: Future implementation research should focus on identifying scalable staffing models and scheduling strategies that balance outreach, capacity, and sustainability when integrating mobile screening services into routine primary care delivery.

Introduction

Breast cancer remains the second most common cancer among women. Screening mammography is a well-established, evidence-based strategy for secondary prevention. However, women living in under-resourced communities are approximately 50% less likely to have received a mammogram within the previous 2 years than women residing in higher income areas [1, 2]. Underserved patients are also more likely to be diagnosed with breast cancer at later stages [3]. Social determinants of health including limited access to screening facilities, transportation and childcare barriers, inflexible work schedules and limited paid time off, high screening costs, cultural norms, and limited understanding of screening importance, are associated with lower mammography screening rates [1, 4–7]. Research suggests that medically underserved patients may benefit most from multimodal, community-based strategies to improve adherence to recommended screening [8]. Accordingly, primary care-based initiatives that address these barriers may reduce disparities in breast cancer detection and mortality [9, 10].

Muli-sector perspectives and external environment

Primary care clinicians play a foundational role in supporting adherence to preventive breast health screening. Clinical recommendations and orders for mammography are critical antecedents in the screening process, and trusted relationships between clinicians and patients may promote screening completion [11]. Despite these influences, mammography utilization remains suboptimal, particularly in underserved communities [12–18].

Evidence supports the use of community-based primary care strategies to reduce patient-level barriers to mammography screening, including reminder systems, scheduling assistance, and patient navigation. Passive reminder interventions, such as letters, postcards, and telephone reminders, represent some of the most effective and feasible intervention strategies implemented in primary care settings to improve mammography completion, with more active approaches involving direct patient outreach and mammography scheduling by clinic staff [19–21]. Active scheduling initiatives have demonstrated substantial improvements in screening completion, particularly in safety-net and women's health clinic settings [22, 23] however, these approaches can be limited by staffing constraints and challenges related to electronic health record interoperability between primary care practices and mammography providers. Patient navigation support has also been implemented in some clinics to provide personalized education and assistance addressing barriers to screening completion [23]. Importantly, culturally relevant primary care-based outreach has shown effectiveness for mammography completion among underserved women [24, 25].

Mobile mammography has emerged as a promising approach for addressing persistent screening barriers in underserved communities. MMUs are portable screening facilities housed in large vehicles that travel to communities in need. By bringing imaging equipment and technologists directly to patients, MMUs reduce barriers related to transportation and cost and often provide free or low-cost screening services [1, 26, 27]. Despite these advantages, prior research has identified challenges to MMU utilization, including limited patient familiarity, inconsistent scheduling and locations, restricted service hours, and vulnerability to inclement weather [1, 4–7]. Routine placement of MMUs onsite at primary care practices has the potential to address both patient- and clinician-level barriers, positioning MMUs as effective partnerships for improving screening access in communities with the greatest need.

Through collaborative planning with clinical, patient, and community partners, we designed and implemented a program to address barriers to mammography screening by embedding mobile mammography services within primary care practices serving medically underserved populations across Western New York State. The purpose of this paper is to describe the implementation

process and evaluate outcomes of this community-based screening intervention, guided by the Practical, Robust Implementation and Sustainability Model (PRISM) for describing the intervention and the Reach, Effectiveness, Adoption, Implementation, and Maintenance (RE-AIM) evaluation framework [28]. This study examines patient ambassador administrative logs, qualitative interviews with clinical partner staff, and anonymous patient-surveys to evaluate real-world implementation outcomes and sustainability associated with a program to increase access to screening mammography via a MMU screening program, facilitated by peer Patient Ambassador prescreening outreach and onsite MMU support.

Methods

Program development and rationale

Partner characteristics

The Patient Voices Network (PVN) is a grassroots group composed of low-income, predominantly Black patients from safety-net practices in Western New York partnering with researchers to develop patient-oriented, community-based interventions to increase cancer screening. Early in their advocacy work, PVN found that community health events reached many people but resulted in very low mobile mammography completion; however, relocating mobile screenings to primary care settings led to a sharp increase in uptake and (e.g. two screened at a Breast Cancer Walk *versus* 78 MMU screenings from walk attendees at primary care practices in the weeks after) [29]. This marked improvement in MMU screening at primary care practices highlighted the potential of primary care-based partnerships and informed the development of the primary care-based MMU screening program presented here.

Sustainability infrastructure

Program partners included the Patient Voices Network, MMU providers, health equity researchers, and local healthcare organizations. Delivering MMU screenings adjacent to primary care practices created an opportunity to develop a sustained, scalable program that consistently brought MMUs to partner clinics. Increasing mammography completion offered mutual benefits by addressing inequities in access for patients while simultaneously supporting quality metrics and performance indicators for participating practices. However, program planning identified a key infrastructure concern: already overburdened and understaffed primary care practices would be unable to absorb additional responsibilities associated with mammography scheduling, front-end check-in, and insurance screening within existing workflows. The peer Patient Ambassador role, hereafter referred to as Ambassadors, was developed to address limited font office resources at primary care sites. Ambassadors were members of PVN community group, paid hourly to help facilitate onsite MMU program outreach, pre-scheduling, and in-person screening. They received training from healthcare practices and university researchers covering breast health and cancer risk, principles of peer support, HIPAA and research ethics, clinical workflows, MMU logistics, and strategies for connecting patients with relevant social services.

Program implementation

The program implementation and following outcomes data catalog the program across Phase 1: initiation (2013–2014) and Phase 2: continuation and expansion (2015–2019). Core features of the breast cancer screening intervention were monthly MMU screening and breast health education with Ambassador liaisons, hosted onsite at participating primary care practices in communities with limited access to mammography services. Implementation strategies emphasized building capacity within primary care clinics, allowing flexibility for site-specific customization, and establishing feedback mechanisms to support continuous improvement. Partner practices agreed upon regular monthly screening schedules, typically beginning with one MMU visit per month per site. Primary care staff were trained to refer patients directly to the MMU based on availability. Although walk-in appointments were permitted, most patients were referred to MMU staff or Ambassadors for prescheduling.

Screening day processes were created to meet practice's unique needs, with Ambassadors utilized to minimize disruption to existing practice workflows. In the most common screening day model, Ambassadors extending clinic capacity by creating a parallel workflow for patients arriving for mammography. The typical MMU visit process included two Ambassadors onsite in a community space at the clinic behind a folding table with handouts, laptop, and decorative banner. Here ambassadors registered patients for MMU by verifying their identification and insurance information, inputting in the electronic health record, providing breast health education handouts, and when prompted by MMU technicians, escorting patients outside to the MMU. Once onboard the unit, X-ray technicians and staff of the MMU took over the screening process and Ambassadors returned to the check-in desk. Ambassadors also served as liaisons between primary care teams and radiology staff by coordinating missing referral orders and enrolling patients in the Cancer Services Program (support for under- or uninsured patients) when necessary.

The Ambassador outreach role evolved in response to practice and patient needs. Ambassadors received lists of patients overdue for mammography screening from partner practices. In some settings, Ambassadors provided navigation-like telephone outreach support to patients overdue for mammography, to offer encouragement, education, and assistance addressing screening-related concerns. In other primary care settings, Ambassadors supported direct appointment scheduling during outreach calls. Ambassadors did not provide clinical advice but addressed personal barriers to screening, including health anxiety, fear, and medical mistrust. When patients described social or resource constraints, Ambassadors connected them to support services through the local chapter of the national 211 database.

Program feedback and refinement

Multiple feedback mechanisms supported program refinement and provided data to inform this outcome evaluation. Ambassadors maintained written log notes during screening days and outreach calls. Anonymous patient satisfaction surveys were collected following MMU screening and reviewed semi-annually to inform program adjustments. Qualitative focus groups and interviews with patients and clinical partners were

conducted over multiple years to capture additional insights into program implementation. Ambassadors also met twice monthly with the research team to review ongoing implementation facilitators and barriers.

As an example of program refinement in response ambassador-patient feedback, a follow-up call feature was added to the program a couple years into Phase 2 to help address patients' not receiving or understanding their mailed test results or appropriate next steps. Several weeks after MMU screening Ambassadors called English-speaking patients; inquiring about receipt of test results, whether they understood the results, and encouraged patients to discuss concerns with their healthcare team. If patients reported needing additional screening, Ambassadors linked them to 3-D imaging centers and resources to assist with transportation.

Outcomes measures, data sources, and analysis

Table 1 summarizes RE-AIM outcome domains and their operationalization while Table 2 catalogues all data sources reviewed in this evaluation and their associated analysis methods. Given the community-based nature of the program, individual-level clinical outcomes and patient demographic data were unavailable to the research team. Program delivery and evaluation were classified as non-human subjects research by the University Institutional Review Board.

Administrative data were collected across Phase 1: initiation (2013–2014) and Phase 2: continuation and expansion (2015–2019). Descriptive and ANOVA analyses were conducted using Microsoft Excel and SPSS version 31. Qualitative data were

Table 1 RE-AIM outcomes measures and operationalization.

Dimension	Outcome of interest	Outcomes measures operationalization	Data source
Reach	Individual-level screening exposure and engagement	1. Total number of mammograms completed via onsite MMU screening 2. Total number of women prescheduled for MMU mammography 3. Total number of patients successfully reached by phone outreach	1. MMU Field Notes 2. Call Logs 3. PA Call Logs
Effectiveness	MMU prescheduling success rate Patient-reported acceptability and perceived access to MMU screening	1. Percentage of outreach phone calls that resulted in MMU screening appointment being scheduled 2. Patient satisfaction with onsite MMU screening services 3. Patient's intention to use the MMU for future mammography screening and to recommend the service to others	1. PA Call Logs 2–3. Patient Satisfaction Surveys
Adoption	Organizational uptake of the MMU program across primary care settings	1. Total number of primary care practices adopting and implementing onsite MMU screening 2. Characteristics of adopting practices, including practice type and setting 3. Practice-reported motivators and barriers influencing adoption of MMU program components	1. MMU Field Notes 2. Program and grant notes 3. Program and grant notes
Implementation	Resource use Facilitators Barriers Adaptations and site-level variation	1. Program adaptations made to support delivery within local practice and patient contexts 2. Descriptive assessment of resources required for implementation, including staffing, coordination, and logistical supports 3. Practice- and program-level facilitators and barriers influencing day-to-day delivery of onsite MMU screening	All: Primary care provider focus groups and MMU Screening Day Logs notes
Maintenance	Organizational-level sustainability of onsite MMU screening	1. Duration of continued MMU program delivery within primary care practices 2. Evidence of adaptation over time to support ongoing implementation and viability	All: MMU Field Notes, Program and grant notes

Table 2 RE-AIM evaluation data sources and analysis approach.

Data source	Description of measure	Analysis method
Primary care provider focus groups	During early implementation (Phase 1), focus groups were conducted with clinical providers at three participating primary care practices ($n = 18$) to assess implementation processes. Discussions elicited provider perspectives on the mobile mammography program's impact on screening uptake, workflow, and patient access, as well as feedback on patient experiences, follow-up processes, and the role and integration of Ambassadors within clinical workflows.	Focus group transcripts from four sessions across three practices were reviewed following a Rapid qualitative analysis approach [30, 31]. Using predefined implementation domains (resource use, facilitators, barriers, site-based adaptations), the first author conducted a thematic synthesis in Excel and tabular formats, with findings reviewed and confirmed by a senior investigator involved in program design and implementation.
MMU screening day logs	Ambassadors maintained deidentified administrative logs from 2014 to 2019 capturing screening date and location, numbers of prescheduled and walk-in patients, and qualitative field notes describing operational issues and patient experiences.	Quantitative log data were summarized descriptively with counts and percentages. Qualitative field notes were reviewed using predefined implementation domains (resource use, facilitators, barriers, and site-based adaptations). The first author conducted a structured rapid qualitative analysis review and thematic synthesis of notes in Excel across program phases, with findings reviewed and confirmed by a senior investigator involved in program implementation [30, 31].
Patient outreach call logs	During outreach calls, Ambassadors documented call outcomes, whether screening appointments were scheduled, and, when applicable, brief qualitative notes.	Quantitative data from call logs were summarized descriptively using counts and percentages.
Patient satisfaction surveys	Anonymous patient experience data were collected using optional, self-administered surveys completed immediately following MMU screening. Survey items used five-point Likert scales to assess level of agreement with statements about program satisfaction and patient intention to reuse and recommend MMU services, with lower scores indicating higher satisfaction. A total of 1531 surveys were collected from 2013 to 2019, with 1426 complete records containing site and year data.	Quantitative survey data were summarized using descriptive statistics, including mean and frequencies for Likert-scale responses. Using structured comparisons across site and year, univariate ANOVA was applied to assess differences in patient-reported experience outcomes, with no significant variation observed. (all $P > .3$).

* Surveys were administered at all participating sites except the rural clinic with limited MMU screening days hosted.

reviewed and grouped in Excel and then summarized into most prevalent findings using Rapid Qualitative Analysis by the first author with PhD level training in qualitative analysis methods [30, 31]. All implementation outcomes were contextualized from focus groups or qualitative notes and confirmed through agreement with the senior author and program lead.

RE-AIM results

RE-AIM evaluation outcomes are presented as Reach, Effectiveness, and then Adoption and Maintenance described together, prior to the implementation outcomes. As shown in Table 1, Reach measured screening volume and outreach contacts. MMU program effectiveness outcomes were prescheduling success rates and patient intention to use and recommend the MMU service to others. Adoption outcomes captured the contexts of participating primary care settings and engagement levels, where Maintenance added organizational-level sustainability indicators: program continuation and adaptability over time. Finally, Implementation was evaluated qualitatively from primary care provider focus groups and Ambassador program logs, detailing, resource use, facilitators, barriers, and program adaptations over time.

Reach

A total of six primary care practices, some with multiple office locations, adopted the program (Table 3). Ambassadors made 10 408 outreach calls to patients across practices, with the number of concurrent partner sites ranging from three to five. During Phase 1, Ambassadors made 1269 calls to women overdue for screening mammography across four healthcare practices and in Phase 2 the program reached more than seven times as many women over 5 years, expanding to six medical practices.

Participating practices hosted a total of 315 onsite MMU screening days over 7 years. These screening days resulted in 6466 mammograms, along with the provision of breast health education. Across the program period, an average of 20 mammograms were completed per hosted screening day. The MMU had a prescheduled capacity of approximately 30 appointments per screening day. However, missed appointments occurred nearly 10 times per day, resulting in 67% attendance rate for prescheduled mammograms. Additional screening slots were also available for walk-in-patients during screening days.

Effectiveness

Outreach call outcomes varied by program year, as shown in Table 4. Phase 1 demonstrated the lowest connection rate, defined as successful contact between an ambassador and a patient during an outreach call. During Phase 2, connection rates ranged from 32% to 43% of all calls. Ambassadors also experienced substantial variability in success when prescheduling women for MMU appointments during outreach calls, with annual prescheduling rates ranging from 15% to 76%.

We collected 1426 complete, anonymous surveys from patients leaving the MMU between 2013 and 2019. Throughout the program, patients reported consistently high satisfaction with their MMU screening experiences (99% satisfied or very satisfied with the service). Univariate ANOVA found no significant differences in patient experience Likert scale ratings by clinical site, program year, or site*year interaction (all $p > .3$). Across program years and sites, 99.2% of patients would recommend the MMU program to others and 99.0% agreed they intended to use the MMU again for future screenings.

Implementation

Implementation outcomes describe how the MMU program functioned operationally within primary care practices and how delivery processes evolved over time. Implementation findings, summarized in Table 5, draw on qualitative focus groups with practices conducted during Phase 1: initiation (2013–2014) and longitudinal review of Ambassador program logs documenting

Table 3 Mobile mammography program reach and maintenance over time by primary care practice.

	Practice site			Program component					
	Type of practice	Location	Active period	Phase 1: Program initiation (2013–2014)			Phase 2: Program continuation and expansion (2015–2019)		
				Outreach calls	MMU days	Screenings completed	Outreach calls	MMU days	Screenings completed
1	University Affiliated	Urban	2013–2019	407	2	71	277	54	877
2	Hospital Clinic	Urban	2013–2019	625	5	71	4176	60	1030
3	FQHC	Urban	2013–2019	183	7	145	2017	164	3914
4	Private	Rural	2014–2015	54	1	33	79	3	68
5	University Affiliated	Urban	2017–2019	–	–	–	1606	14	191
6	Private	Suburban	2018–2019	–	–	–	984	5	66
Total				**1269**	**15**	**320**	**9139**	**300**	**6146**

This table shows primary care practice site characteristics for those implementing the MMU screening program as well as outcome data for program initiation (2013–2014) and continuation/expansion phases (2015–2019). These data were derived from Patient Ambassadors' field notes administrative records from mobile mammography unit (MMU) onsite screening days at partner medical practices, as well as logs kept during outreach phone calls to patients. Only data with both date of occurrence and site information were included in these reports.

Table 4 Annual patient ambassador's outreach call outcomes for mammography screening program 2013–2019.

Year of outreach	Total calls	Spoke to patient	Connection rate	Number of MMU screenings pre-scheduled	Telephone pre-scheduling success rate
2013	880	201	22.8%	78	38.8%
2014	389	188	48.3%	143	76.1%
2015	1642	672	40.9%	350	52.1%
2016	298	129	43.3%	40	31.0%
2017	818	308	37.7%	65	21.1%
2018	2266	799	35.3%	160	20.0%
2019	2815	907	32.2%	134	14.8%

Table 5 Summary of implementation findings across program initiation and continuation phases by data source.

		Program initiation (2013–2014)	**Program continuation (2015–2019)**
	MMU daily program logs (*n* = 8)	**Provider focus groups (*n* = 18)**	**MMU daily program logs (*n* = 158)**
Resource use:	**Coordination:** rescheduling women that did not show up for MMU screening represents an ongoing clinical staff burden	**Time savings for providers:** Ambassadors manage scheduling, reminders, and logistics, freeing clinicians to focus on patient care. *Site 3 Provider: "We don't have time to call and remind patients … she's doing all of that—call, schedule—and it just makes things easier for us."* **Operational efficiency:** MMU on site improved communication of results with providers, reduced lag time for follow-up, improving care continuity. **Volunteer dependency:** Heavy reliance on ambassadors as part time volunteers creates sustainability concerns and risk of burnout.	**High no-show rate and scheduling issues:** Frequent cancellations, reschedules, and patient no-shows, limit utility of available appointments. **Staffing needs:** Managing patient flow and coordinating prescription for the scan require significant coordination with staff time. Two Ambassadors are needed at the check-in desk to maintain patient flow and assist with paperwork. **Supplies and amenities:** ongoing confusion over which partner is responsible for providing and stocking MMU day supplies and patient paperwork
Facilitators:	**Patient familiarity:** Most familiar with mammograms, likely reducing apprehension about trying new onsite MMU program **Positive workflow:** Workflow was described as "pleasant," according to ambassadors indicating smooth integration of the program into clinic operations.	**Convenience and accessibility:** MMU at familiar clinical locations reduces transportation barriers and allow same-day screening during regular primary care visits. **Patient support through ambassadors:** Ambassadors guide patients, provide screening education, and provide patient social support. Especially effective when culturally aligned or sharing personal experiences. **Workflow integration:** Ambassadors handling patient scheduling minimizes clinical burden and ensures smooth clinic operations. *Site 2 Provider: "… when the mammogram bus [is here] … they handle everything and that's good for us."*	**Positive staff engagement:** Notes indicate smooth operations and strong staff commitment. **Patient comfort and education:** Ambassadors reassured first-time patients and explained the importance of mammograms; some patients expressed appreciation for the onsite screening experience. **Community connection:** Interactions with patients and notes suggesting additional community outreach show program engagement beyond the clinic.
Barriers:	**Accessibility challenges**: Concern about assisting patients on and off the MMU suggests mobility difficulties for some individuals; MMU not wheelchair accessible therefore some patients will need alternative screening options.	**Limited capacity:** Not enough mobile unit days and ambassadors to meet demand; long waiting lists for MMU persist. *Two Providers at Site 3: "There's already a waiting list … we constantly have patients."*	**Prescription and documentation problems:** Missing or expired mammogram orders; miscommunication about script process; patients arriving without required documents (identification, insurance card) **Accessibility challenges**: MMU

(continued)

Table 5 (continued)

		Program initiation (2013–2014)	Program continuation (2015–2019)
	MMU daily program logs (*n* = 8)	**Provider focus groups (*n* = 18)**	**MMU daily program logs (*n* = 158)**
		"I wish we could increase the number of days … we can't right now." **Space constraints:** Lack of dedicated space for Ambassadors in the clinic, often relegated to set up check in desk in hallways. **Ambassador recruitment challenges:** Difficulty recruiting new ambassadors due to offering limited work hours per month; risk of burnout among current team. **Technology and training gaps:** Ambassadors lack EMR skills, limiting independence and efficiency.	is not accessible to wheelchairs; difficulty navigating stairs for elderly; advance communication about MMU accessibility is needed. **Technical and operational issues:** MMU breakdowns, clinic closures, and lack of space for ambassadors to check-in MMU patients; causing unknowns for patients and staff burden for rescheduling patient screenings
Site based adaptations:	*None presented*	**Cultural and language needs:** Refugee-focused clinical sites rely heavily on interpreters and culturally aligned ambassadors, however there is limited diversity within Ambassador team. **Outreach approaches:** Some sites depend on Ambassador outreach calls, but others rely on word-of-mouth, self-referral or clinical staff scheduling, possibly due to differing language needs. *Site 3 Provider: "People see the bus and request it … it's a ripple effect."* **Infrastructure and workflow:** Variability in Ambassadors integration with EMR systems; some sites request ambassador training for greater independence. **Capacity and demand:** Certain sites report urgent need for more regularly scheduled MMU days and Ambassadors due to high demand, while others view the program as a community enhancement rather than essential service. *Site 1 Provider: "I don't know that it increased screening … but people prefer the [MMU] bus."*	**Cultural and language needs:** Multiple languages noted; need for interpreters and bilingual educational materials. **Scheduling adjustments:** Some sites began directly scheduling their own patients for MMU at their annual appointments, while other practices relied solely on patient ambassador outreach calls to pre-schedule women for MMU. Modified appointment blocks to avoid lunch break cnflicts; stopped late-day scheduling; some sites increased reminder calls and outreach for screening days. **Communication improvements:** Some suggested providing clearer check-in location instructions and MMU accessibility information in advance. Some asked for registration forms in large print to help with vision impairment.

Implementation findings derived from multiple qualitative data sources collected across program phases. This included focus group implementing primary care providers (*n* = 18) across urban sites 1–3 as well as longitudinal MMU program logs captured early workflow challenges (2013–2014; *n* = 16 completed logs; *n* = 8 with qualitative notes) and contextual considerations during program continuation (2015–2019; *n* = 298 completed logs; *n* = 158 with qualitative field notes).

MMU screening day activities Phase 2: continuation and expansion (2015–2019).

Resource use: Prior to program implementation, providers identified pre-scheduling and MMU patient check-in demands as primary workflow constraints. Early program implementation was facilitated by the delegation of prescheduling, patient reminders, and onsite MMU logistics to Ambassadors, which reduced administrative burden on clinical staff and enabled providers to focus on patient care. These findings indicate that redistribution of administrative tasks was a key element to supporting feasibility during early program adoption.

Facilitators: Ambassadors managed scheduling and logistics, minimizing clinical burden and contributing to smooth workflow integration, while positive staff engagement and community connections further supported program delivery. The convenience of offering MMU screening at familiar primary care locations reduced transportation barriers and supported screening during routine visits. Implementation findings reflected operational refinements aimed at improving reliability and efficiency of onsite screening delivery. Ambassador program logs documented improvements in patient flow and coordination, including greater use of pre-completed paperwork, standardized check-in procedures, and modifications to appointment blocks to reduce missed screenings. The onsite MMU design also improved communication between radiology teams and primary care providers, reducing lag time for follow-up and supporting continuity of care.

Barriers: At the same time, persistent implementation barriers were observed. Limited MMU availability, technology breakdowns, and Ambassador staffing constrained capacity relative to demand, high patient no-show rates required ongoing rescheduling efforts, and managing patient flow frequently necessitated two Ambassadors onsite to maintain efficient operations. Reliance on Ambassadors being paid from grants, rather than from practice or MMU service funds, introduced sustainability concerns for the ambassador staffing model once grants ended. Additional barriers included challenges having dedicated workspace for Ambassadors, ambiguity in responsibility for office supplies and documentation, and accessibility limitations of the MMU for patients with mobility impairments.

Site based-adaptations: Implementation processes varied across practice settings in response to local context. Sites serving diverse linguistic and cultural populations relied more heavily on clinical/staff-interpreters, bilingual materials, and Ambassadors receiving additional training on the nuanced needs of particular immigrant groups (e.g. patients arriving with a male escort). Variability was also observed in Ambassador integration with scheduling systems and electronic medical records, as well as in outreach strategies and patient reminder approaches.

Adoption and maintenance

Site adoption and program uptake: The breast health awareness and screening partnership expanded over time, with a total of six primary care practices, some with multiple office locations, adopting the program with Ambassador support. Most participating practices were urban sites (67%), with one suburban and one rural practice. Initial adoption included three urban primary care practices, followed by a rural practice that piloted the program in 2014. During Phase 2, a university-affiliated urban practice and a suburban private practice each implemented the onsite screening program for approximately 2 years. An overview of participating practices, partnership duration, and levels of engagement in Ambassador-supported breast health promotion activities over time is presented in Table 2. All but one practice ended participation with the onset of the COVID-19 pandemic, with most maintaining the program for approximately 7 years in total. The private rural practice did not maintain participation beyond 2015 due to challenges scheduling the MMU on-site and coordinating logistics. The federally qualified health center sustained all program components beyond 2019.

Across all partner sites, Ambassadors facilitated onsite MMU screening days and conducted patient outreach phone calls. However, the intent of outreach activities, such as patient education versus direct appointment scheduling, varied by practice and was not consistently tracked over time. During Phase 1, three to four Ambassadors were cross-trained to support multiple practices. As the number of participating sites increased, Ambassador staffing levels remained relatively stable, with annual staffing ranging from three to five Ambassadors, reflecting sustained engagement over time.

Motivators and barriers to adoption: Key motivators for adopting MMU based screening within primary care practices included the need to improve breast cancer screening rates, particularly among sites serving high proportions of Medicaid insured patients facing substantial barriers to care. These barriers included limited access to screening services in both urban and rural settings, low health literacy, transportation challenges, and broader socioeconomic constraints. The MMU supported participating practices in meeting quality metric incentives and aligning with patient centered medical home models. Early adopting practices, particularly in urban settings, served predominately racial and ethnic minority patients, including Black or African American patients, and, in the case of a federally qualified health center, many new Americans and individuals with limited English proficiency. Adoption was facilitated by engaged clinic leadership and practice champions who supported integration of MMU screening into routine workflows. Leadership support enabled scheduling consistency, coordination across staff roles, and prioritization of MMU screening amid competing clinical demands.

Barriers to sustained adoption included limited perceived impact among sites that hosted the MMU infrequently, such as on a quarterly basis, which may have reduced integration with routine clinical workflows and limited opportunities to build patient familiarity and trust. MMU capacity plateaued at approximately 30 screenings per day causing screening rates to plateau unless additional days were added. Practices that hosted the MMU monthly were better positioned to meet patient demand and offer consistent access to screening services over time.

Maintenance and sustainability over time: Program maintenance varied across primary care sites and was closely linked to differences in practice preferences and implementation intensity. Although all participating sites engaged in both patient outreach calls and onsite MMU screening, sites that achieved greater screening success were more likely to sustain program participation over time. For example, one urban hospital-affiliated primary care clinic prioritized Ambassador led outreach calls focused on

identifying patient questions or support needs related to mammography, rather than emphasizing direct appointment prescheduling. Ambassadors conducted the highest volume of outreach calls on behalf of this site, accounting for 46% of all 10 408 patient calls, yet the site represented only 17% of 6466 completed mammograms.

In contrast, a multilingual urban federally qualified health center accounted for approximately 21% of outreach calls, totaling about 2200 calls, but represented 63% of all 6466 completed onsite MMU mammograms during the program period. This site prioritized consistent, predictable onsite screening days and clinic-based direct MMU prescheduling due to needing to schedule women with limited English literacy since all Ambassadors were English-speaking only. Across sites, consistent scheduling, such as hosting MMU screening on the same day each month and integrating direct appointment prescheduling within clinic workflows, appeared to be a stronger facilitator of mammogram completion than patient outreach calls alone.

Discussion

This study examined the implementation of a mobile mammography program, with MMU's coming onsite adjacent to primary care practices serving medically underserved communities, using PRISM to contextualize program and implementation characteristics and REAIM to evaluate program outcomes. Across REAIM domains, findings demonstrate that consistent integration of mobile mammography into primary care workflows, supported by peer Patient Ambassadors and clinic leadership, can increase access to screening while revealing important tradeoffs related to staffing, scheduling intensity, and sustainability.

A central finding across Adoption, Implementation, and Maintenance outcomes was the importance of predictable onsite MMU scheduling. Practices that hosted MMU screening days at consistent intervals and integrated direct appointment prescheduling into clinic workflows achieved higher screening completion and were more likely to sustain participation over time. In contrast, reliance on outreach alone, without predictable onsite availability, was associated with lower screening yield despite high outreach volume. These findings suggest that implementation intensity, particularly scheduling regularity, may be a more critical driver of sustained screening access than outreach volume alone, especially in resource constrained settings. Prior work similarly suggests that use of proactive scheduling strategies can maximize utilization of MMU services and avoid inefficiencies associated with episodic or event-based MMU deployment [4].

The success of this collaborative, community-driven program was in part due to responsiveness to patient and organizational context, key PRISM domains for implementation success. For example, in this program, post screening follow-up calls were added to better support patients and as a direct result of Ambassador-patient interactions identifying gaps in communication and understanding about screening results and patients' appropriate next steps. This adaptation was also in response to prior research suggesting that, particularly among underserved populations, lack of post screening communication when receiving mailed mammography results can contribute to patient anxiety and disengagement [32, 33]. Although reach of follow-up calls was limited, particularly in linguistically diverse settings, this adaptation highlights how implementation processes were modified in response to patient identified needs rather than predefined program protocols.

Implementation findings illustrate the dual role of Ambassadors as both facilitators of access and a critical source of operational capacity. By managing prescheduling, reminders, check-in workflows, and patient education, Ambassadors reduced administrative burden on primary care staff and enabled practices with limited capacity to host extended onsite MMU screening days. At the same time, reliance on Ambassadors, paid from grant support, introduced constraints related to staffing stability. Similar studies have shown that peer or navigator roles can enhance screening uptake but require sustained investment to ensure scalability and long-term integration within clinical systems [34]. These tradeoffs underscore the importance of aligning program design with realistic staffing and infrastructure expectations (like technology access and training) when embedding community-based screening models within healthcare delivery systems.

Equity considerations were evident throughout implementation and site-specific adaptations. Practices serving refugee and immigrant populations required additional linguistic and cultural resources, including interpreters, bilingual materials, and cross-culturally trained Ambassadors, reinforcing the importance of tailoring implementation strategies to local patient populations. These findings align with prior work demonstrating that community trust and culturally responsive engagement are critical for equitable delivery of preventive services in underserved settings [34, 35]. However, much like other programs implementing MMU screenings, accessibility limitations of the MMU itself, including lack of wheelchair access and mobility challenges for some patients, point to structural barriers that mobile screening programs must address to avoid unintentionally excluding high-need populations [7, 26].

Several limitations should be considered when interpreting these findings. This evaluation relied on administrative and qualitative data collected for program operations rather than controlled research measures, resulting in incomplete tracking of some outcomes and variability in documentation over time. Individual-level clinical outcomes and demographic data were unavailable, limiting assessment of differential reach or effectiveness across subpopulations. Additionally, MMU operations were limited to weekday daytime hours, which may have constrained access for individuals with inflexible work schedules, as prior studies suggest evening and weekend screening hours may increase uptake [36]. Finally, program sustainability remained dependent on grant funding and university support, highlighting ongoing challenges to financing peer supported, community-embedded- screening models at scale [37].

Conclusions

Together, these findings highlight how sustained collaboration among community members, Patient Ambassadors, primary care practices, and researchers can strengthen health systems to improve access to screening mammography and reduce

persistent disparities. Peer based engagement approaches, integrated across program design and delivery, supported implementation within routine care settings and helped align services with patient and community needs. By partnering with patients as contributors to care delivery, practices can develop localized systems that enhance preventive service access while remaining responsive to resource constraints and context.

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Conflicts of interest

None declared.

Human rights

This article does not contain any studies involving human participants performed by any of the authors. Program delivery and evaluation activities were determined to be non-human subjects research by the University's Institutional Review Board.

Informed consent

Informed consent was not required because this study involved program evaluation activities determined to be non-human subjects research by the Institutional Review Board.

Welfare of animals

This article does not contain any studies with animals performed by any of the authors.

AI disclosure

Microsoft Co-Pilot artificial intelligence-assisted tools were used for limited editorial support, including grammar and clarity improvements. The authors independently developed and verified all scientific content and are fully responsible for the manuscript.

Transparency statement

Study registration

This study design was not preregistered.

Analytic plan pre-registration

This analytic plan was not preregistered.

Analytic code availability

Data and analytic code are not publicly available due to the contextual, administrative, and qualitative nature of the program data and to protect the privacy of participating organizations and individuals.

Materials availability

There are no study related materials to share, though the autors are open to supportive discussion about similar program implementation.

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